

Solution Requirements

Date	29 June
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	System shall provide multi-factor authentication (MFA) via email token or secure credential keys for authorized operations personnel to securely access dashboards.	High
2	Authorization levels	Role-based access controls defining specific view permissions for Disaster Management Officers versus configuration settings restricted to Admin levels.	High
3	External interfaces	Connects via scheduled API requests to structural meteorology telemetry endpoints and real-time regional river stream gauge databases.	High
4	Transactions processing	Handles persistent event log streaming updates, high-throughput message bus operations, and transaction rollback metrics across local nodes.	Medium
5	Reporting	Generates downloadable incident summaries, maps showing warning radii, dynamic Flask dashboard data visualization cards, and telemetry summaries.	High
6	Business rules, etc.	Triggers high-alert escalations automatically when precipitation metrics hit threshold limits or ML classification scores exceed predefined tolerances.	High
7	Compliance to laws or regulations	Ensures compliance with standard data encryption norms, regional disaster reporting frameworks, and emergency information privacy laws.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/ Medium/ Low)
8	Other	Provides an automated heartbeat ping tracker tool to verify physical network availability at local stream monitoring checkpoints.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Ensures low latency reporting updates and rapid generation of warning charts for users.	Page loads and map renders in < 2 seconds
2	Scalability	Handles variable data payload surges during flash monsoon environments across regions.	Supports up to 10,000 concurrent open user sessions
3	Security & Data Privacy	Protects critical command feeds, alert relays, and private credentials across nodes.	End-to-end AES-256 transmission encryption
4	Reliability & Availability	Maintains continuous operation without service dropouts during critical flood windows.	99.9% uptime per calendar month
5	Usability & Accessibility	Guarantees intuitive interface structures for stress-filled emergency command teams.	Complies with WCAG 2.1 AA digital standards
6	Other	Ensures code microservices can be redeployed across dynamic backup hosting containers.	Containerized portability with sub-10 min setup time